INCH-POUND

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15 April 1971

MILITARY SPECIFICATION

RECEIVER, RADAR R-1475/VPS-2: 10548338

This **specification is approved for use by the U. S. Army Armament, Munitions and Chemical Command, and is available for use by all Departments and Agencies of the Department of Defense.**

10 SCOPE

1.1 Scope. This specification covers the general requirements and tests for Receiver, Radar R-1475/VPS-2, hereinafter referred to as the receiver, which is a unit of Radar Set AN/VPS-2 (see 6.1).

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards and handbooks. The following specifications, standards and handbooks form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2).

SPECIFICATIONS

MILITARY

MIL-F-13926	Fire Control Materiel, Manufacture and Inspection, General Specification for
MIL-I-45607	Inspection Equipment, Acquisition, Maintenance and Disposition of
MIL-R-50651	- Radar Set AN/VPS-2: 10549058
MIL-O-50665	Oscillator Doppler Assembly

Beneficial comments (recommendations, **additions**) and any pertinent data which maybe of use in improving this document should be addressed to: Commander, U.S. Army ARDEC, ATTN: SMCAR-BAC-S, Picatinny Arsenal, New Jersey 07806-5000 by using self-addressed Standardization **Document Improvement Proposal (DD Form 1426)** appearing at the end of thid socument or by letter.

AMSC N/A FSC 1285

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MIL-C-50675 - Circuit Assembly AGC: 10549228
MIL-G-50676 - Gate I-F: 10548649
MIL-D-50677 - Detector Doppler: 10548509
MIL-G-50678 - Gate Multiple Circuit: 10548325

STANDARDS

MILITARY

MIL-STD-109 - Quality Assurance Terms and Definitions
MIL-STD-810 - Environmental Test Methods and Engineering Guidelines
MIL-STD-2000 - Standard Requirements for soldering **Electrical and Electronic Assemblies**

(Unless otherwise indicated, copies of federal and military specifications, standards, and handbooks are available from the Standardization Documents Order Desk, Building 4D, 700 Robbins Avenue, Philadelphia, PA 19111-5094.)

2.1.2 Other Government documents, drawings and publications. The following other Government documents, drawings and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues are those cited in the solicitation.

DRAWINGS (See 6.4)

U. S. ARMY ARMAMENT RESEARCH, DEVELOPMENT AND ENGINEERING CENTER (ARDEC)

10548158 - Oscillator
10548166 - Balanced Mixer
10548183 - Hybrid Junction
10548338 - Receiver, Radar R-1475/VPS-2

PACKAGING DATA SHEET

PS10548338 - Packaging of Receiver

(Copies of drawings, publications and other Government ddocuments **required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)**

2.2 **Order of precedence. In the event of conflict between the text of this document and the references cited herein , the text of** this document takes precedence. Nothing in this document, however, supersedes applicable laws **and regulations unless a specific exemption has been obtained.**

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3. REQUIREMENTS

3.1 Materials. Materials shall be in accordance with the applicable drawings and specifications.

3.2 Components and assemblies. The components and assemblies shall comply with all requirements specified on Drawing 10548338, all associated drawings, and with all requirements specified in applicable specifications and standards. The subassemblies of the receiver shall have been inspected and accepted in accordance with their respective drawing or specification prior to their installation into the receiver. The following subassemblies are covered by the indicated specifications:

Unit No.	Name	Specification
3A11	Oscillator Doppler Assembly	MIL-O-50665
3A2	Circuit Assembly AGC!	MIL-C-50675
3A9	Gate I-F Assembly	MIL-G-50676
3A6	Detector Doppler Assembly	MIL-D-50677
3A5	Gate Multiple Circuits Assembly	MIL-G-50678

Where applicable, the receiver shall meet the requirements of MIL-F-13926 as follows:

a. Dimensions and tolerances.
b. Inorganic protective surface finishings.
c. Part identification and marking.
d. workmanship.

3.3 Environmental service conditions.

3.3.1 Vibration. The receiver shall not suffer damage and shall meet the requirements of 3.4 and 3.5 after exposure to sinusoidal vibrations, in each of three mutually perpendicular axes, as follows:

Frequency Range	Amplitude
10 to 15 hertz	0.06 inches double amplitude
15 to 25 hertz	0.04 inches double amplitude
25 to 55 hertz	0.02 inches double amplitude

The frequency of vibration shall be swept logarithmically. The cycling time shall be 120 minutes per axis. The sweep rate shall be 3.75 minutes ± 10% per decade. The 10-55-10 hertz sweep time is 3 minutes 45 seconds ± 10%.

3.3.2 Shock. The receiver shall not suffer damage and shall meet the requirements of 3.4 and 3.5 after exposure to a total of 18 impulse shocks consisting of three shock impulses of 15 g's ± 10%, 11 milliseconds ± 10%, half-sine wave applied in each direction of 3 mutually perpendicular axes.

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3.3.3 Altitude.

3.303.2 Operating. The receiver shall meet the requirements of 3.4 and 3.5 during and after exposure to barometric pressure encountered at 10,000 feet above sea level (20.6 inches of Hg).

3.3.3.2 Non-operating. The receiver shall meet the requirements of 3.4 and 3.5 after exposure for not less than 10 minutes to barometric pressure encountered at 40,000 feet above sea level (5. 5 inches of Hg).

3.3.4 Temperature.

3.3.4.1 Operating. The receiver shall meet the requirements of 3.4 and 3.5 when stabilized thermally at any temperature between -40°F and +155°F.

3.3.4.2 Non-operating (storage). The receiver shall not suffer permanent damage when subjected to the constant temperature extremes of -60°F for a period of 24 hours and +160°F for a period of 4 hours (after temperature stabilization), and shall meet the requirements of 3.4 and 3.5 when returned to room ambient.

3.3.5 Humidity (non-operating) The receiver shall meet the requirements of 3.4 and 3.5 after exposure to an atmosphere with a relative humidity in excess of 95 percent for a period of 10 days.

3.4 Electrical. Each conductor shall be electrically continuous as specified on the schematic diagram of the receiver.

3.5 Performance. The receiver shall meet the following performance requirements when supplied with input voltages as shown in Table I (Input voltages) and input signals as shown In Table II (Input signals) .

TABLE 1. Input voltages.

CONNECTOR	APPLIED VOLTAGES
J3-E	+15 Vdc, 1% reg.
J3-F	-15 Vdc, 1% reg.
J3-P	± 15 Vdc, return
J3-G	+28 Vdc for range cal test
J3-A	+25 Vdc, 1% reg.
J3-B	+25 Vdc, return
J4-E	+15 Vdc, 5% reg.

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TABLE I. Input voltages (continued).

CONNECTOR	APPLIED VOLTAGES
J4-M	± l5 Vdc, return
J4-F	-15 Vdc, 5% reg.
J4-A	+28 Vdc return
J4-H	+28 Vdc for clutter lock-on test

TABLE II. Input signals.

CONNECTOR	APPLIED SIGNAL
J2-B	Pulse No. 2, Fig. 1
J2-E	Pulse No. 1, Fig. 1
J2-D	Pulse No. 3, Fig. 1
J5-A	60 MHz at +2o dBm

3.5.1 Multiple gate signals.

3.5.1.1 Early Gate RF burst. An Early Gate burst shall be provided at 3P42 which, when connected to a 50-ohm termination shall exhibit the following parameters:

a. Synchronous with Early/Late Trigger.
b. Pulse repetition frequency the same as Early/Late Trigger.
c* Amplitude not less than 550 millivolts (mV) peak-to-peak and not more than 850 mV peak-to-peak.

3.5.1.2 Late Gate RF burst. A Late Gate burst shall be provided at 3P42 immediately following each Early Gate burst. The "Late Gate RF signal shall be 180° out of phase with the Early Gate RF signal. The Late Gate burst amplitude shall be not less than 550 mV peak-to-peak and not more than 850 mV peak-to-peak.

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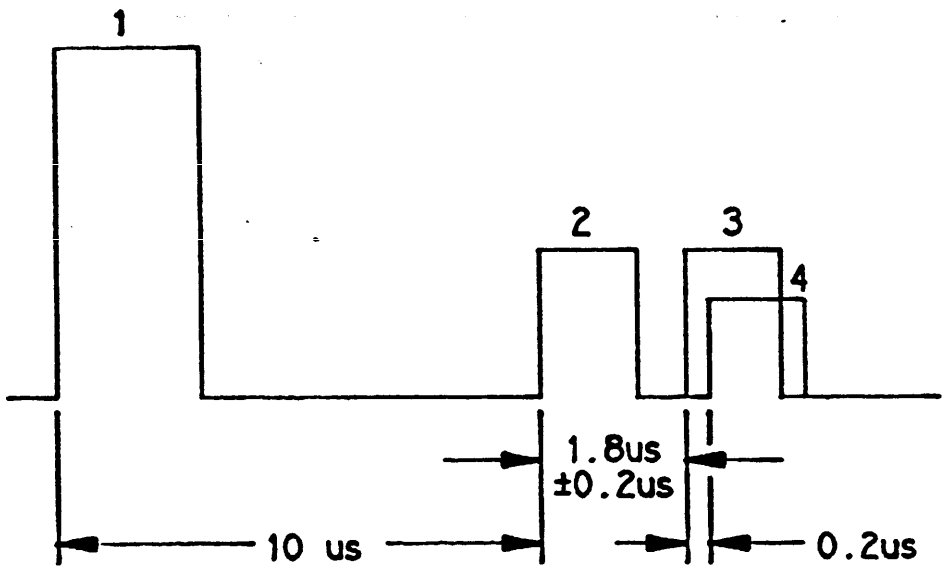


FIGURE 1. Pulse timing chart.

Pulse Parameters:

1.	Mod Trigger	Amplitude:	+15 ± 0.5 V
		Width:	0.5 ± 0.15 us
		PRF:	20 ± 0.1 kHz
2.	Pre-Range Trigger	Amplitude:	+3 ± 0.1 V
		Width:	1 ± 0.2 us
		PRF:	Same as No. 1
3.	E/L Trigger	Amplitude:	+3 ± 0.1 V
		Width:	0.5 ± 0.1 us
		PRF:	Same as No. 1
4.	Target Trigger	Amplitude:	+0.5 ± 0.1 V
		Width:	0.5 ± 0.1 us
		PRF:	Same as No. 1

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In addition, the difference in amplitude between the Early Gate burst and the Late Gate burst shall not exceed 130 mV peak-to-peak.

3.5.1.3 Sum Gate RF burst. A Sum Gate RF burst shall be provided at 3P45 which, when connected to a 50 ohm termination shall exhibit the following parameters:

a. Synchronous with Early/Late trigger. "
b. Pulse repetition frequency the same as Early/Late trigger.
c. Amplitude not less than 616 mV peak-to-peak and not more than 1.18 volts (V) peak-to-peak.
d. The amplitude of the algebraic sum of the Early Gate (measured at 3P42) and Sum Gate signals during the Early Gate period shall be no greater than one-half the amplitude of the Sum Gate signal. The amplitude of the algebraic sum of the Late Gate (measured at 3P42) and inverted Sum Gate signals during the Late Gate period shall be no greater than one-half the amplitude of the Sum Gate signal.

The Sum burst RF signal shall be in phase with the Late Burst RF signal.

3.5.2 AGC action. AGC voltage shall be provided at 3A2J5 and 3J1-E. With the input signal as shown below, the AGC voltage shall be adjustable to not less than 6.0 volts DC (Vdc) and not more than 7.0 Vdc using the video gain adjustment on the 3A1 i-f amplifier to obtain the desired voltage. Input signal characteristics are:

a. Frequency: 60 MHz
b. Width: 0.5 us.
c. PRF : 20 KHz*
d. Amplitude: 90 mV peak-to-peak.

3.5.2.1 Input video signal pulse. An input video signal pulse of not less than 525 mV and no more than 625 mV at 3A2J1 shall produce a +5 V pulse level at 3A2J5. At environmental extremes, the video pulse shall be no less than 412 mV nor greater than 882 mV.

3.5.3 Doppler oscillator. The doppler oscillator output shall be adjustable to obtain 70 mV peak-to-peak at 3A12J2.

3.5.4 Range calibrate function. With a 60 megahertz (MHz) rf level of -14 decibel referred to one milliwatt (dBm) input at 3J5-G, with 28 Vdc at 3J4-H, and relay, 3K1, energized, the sum channel output at 3J2-F shall not be less than 150 mV root-mean-square (rms).

3.5.5 I-f amplifier gain. The i-f amplifier gain from 3A1J1 to 3A1J2 and from 3A1J1 to 3A1J3 shall be 38 ± 2 decibel (dB). The outputs at 3A1J2 and 3A1J3 shall not differ by more than 1 dB.

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3.5.6 **I-f amplifier dynamic range.** The i-f amplifier shall not compress more than 1 dB at either i-f output with an input signal level of -30 dBm.

3.5.7 Sum and Difference channels phasing.

3.5.7.1 Sum and Early Gate phasing. With 59.941 MHz i-f target signal input at 3J1-D and with the target signal centered in the Early Gate, the audio output from the Sum and Difference channels shall have a phase difference of 180° ± 3°.

3.5.7.2 Sum and Late Gate phasing. With 59.941 MHz i-f target signal input at 3J1-D and with the target signal centered In the Late Gate, the audio output from the Sum and Difference channels shall have 0° ± 3° phase relationship.

3.5.8 Minimum discernible signal.

3.5.8.1 Minimum discernible signal, sum channel. with 59.941 MHz i-f target signal applied to 3J1-D, at a level of -75 dBm, and with the target centered in the Sum Gate, the output power at 3J2-F shall not be less than twice the power output with no i-f signal input at 3J1-D.

3.5.8.2 Minimum discernible signal, difference channel.

3.5.8.2.1 Minimum discernible signal, target in Early Gate. With 59.941 MHz i-f target signal Input applied to 3J1-D, at a level of -75 dBm, and with the target signal centered in the Early Gate, the output power at 3J2-G shall not be less than twice the power output with no i-f signal input at 3J1-D.

3.5.8.2.2 Minimum discernible signal, target in Late Gate. With 59.941 MHz i-f target signal input applied to 3J1-D, at a level of -75 dBm, and with the target signal centered in the Late Gate, the output power at 3J2-G shall not be less than twice the power output with no i-f signal input at 3J1-D.

3.5.9 60 MHz output. A 60 MHz CW stalo signal shall be provialed at J5-E at not less than +13 dBm power level when terminated in 50 ohms.

3.5.10 Mod trigger output. A mod trigger output pulse shall be provided at 3J5-F with the following parameters when terminated in a 50 ohm load:

a. Amplitude: +15 ± 0.5 v peak.
b. Pulse width at 50% points: 0.5 ± 0.15 us*
c. Rise time: Not more than 35 ns.
d. PRF : 20.0 + 0.1 KHz.

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3.5.11 System capability. The receiver, after having met the requirements of 3.5.1 through 3.5.1o, shall meet the following requirements of MIL-R-50651:

a. Mid-range cal function (Para 3.5.7)
b. Subclutter visibility (Para 3.5.9.8)
c. AGC characteristics (Para 3.5.9.9) . . . . . .

3.6 First article. When specified in the contract or purchase order, sample shall be subjected to first article inspection in accordance with the technical provisions herein (see 4.3).

3.7 Soldering. The receiver shall be soldered per general requirements an Tasks A, C and D of MIL-STD-2000.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements (inspections and tests) as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items shall meet all requirements of Sections 3 and 5. The inspections set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of assuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling in quality conformance does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to acceptance of defective material.

4.1.2 General Provisions. The component and subassembly inspection requirements of MIL-F-13926 form a part of the quality assurance Provisions of this specification. Reference shall be made to MIL-STD-109 to define quality assurance terms used herein.

4.2 classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.3).
b. Quality conformance inspection (see 4.4).
c. Special sampling inspection (see 4.5).

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4.2.1 Inspection conditions. Unless otherwise specified, all examinations and tests shall be conducted at the following standard ambient conditions:

a.	Temperature	+50° to +90°F.
b.	Relative humidity	90% RH max.
c.	Pressure -	28 to 32 inches of Hg.

4.3 First article inspection.

4.3.1 First article submission. The contractor shall submit a first article sample as designated by the contracting officer consisting of three (3) receivers for inspection in accordance with 4.3.2.

4.3.2 Inspections to be performed. First article assemblies, components and test specimens may be subjected by the Government to any or all of the examinations and tests specified in Table III (Functional tests) and Table IV (Special sampling) herein and to any or all requirements of the applicable drawings.

4.3.3 Rejection. If any assembly, component or test specimen fails to comply with any of the applicable requirements, the first article sample shall be rejected. The Government reserves the right to terminate its inspection upon any failure of any assembly, component or test specimen to comply with any of the stated requirements.

4.4 Quality conformance inspection.

4.4.1 Inspection lot formation. The term "inspection lot" is defined as a homogeneous collection of units of product from which a representative sample is drawn or which is inspected 100 percent to determine conformance with applicable requirements. Units of product selected for inspection shall represent only the inspection lot from which they are drawn and shall not be construed to represent any prior or subsequent quantities presented for inspection. Homogeneity shall be considered to exist provided the inspection lot has been produced by one manufacturer, in one unchanged process, using the same materials and methods, in accordance with the same drawings, same drawing revisions, same specifications and same specification revisions, and complies with the provisions for submission of product. All material submitted for inspection in accordance with this specification shall comply with the homogeneity criteria specified herein, regardless of the type of inspection procedure which is being applied to determine conformance with requirements.

4.4.2 Examinations and tests.

4.4.2.1 Components, assemblies and subassemblies. All components, assemblies and subassemblies shall be inspected in

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accordance with the inspection provisions contained in this specification and the Quality Assurance Provisions (QAP) listed in the Technical Data Package (TDP). In the absence of QAPs, the applicable quality assurance provisions of MIL-F-13926 shall apply.

4.4.2.2 Final acceptance inspection. Subsequent to first article approval, examinations and tests related to Section 3 herein shall be performed on a single defect (individual characteristic) basis in accordance with the sampling plans specified In 4.4.2.3 herein. Examination and test for packaging and marking shall be in accordance with Section 5 herein. The tabulated classification of defects in 4.4.2.3 shall constitute the minimum inspection to be performed by the contractor after first article approval and prior to Government acceptance or rejection by item or lot.

4.4.2.3 Functional tests. The tests in Table 111 shall be performed on a 100% basis.

TABLE III. Functional tests (100% inspection).

CHARACTERISTIC	REQUIREMENT	TEST PROCEDURE
Critical:	None	N/A
Major:		
101. Electrical	3*4	4.7.2
102. Performance	3*5	4.7.3
Minor:	NONE	N/A

4.5 Special sampling. One receiver out of the first 50 produced and one out of each 100 produced thereafter shall be selected at random. The sample shall have met the requirements and tests of Table III and shall then meet the requirements and tests in Table IV. Unless otherwise specified in the contract, the items shall be returned to the lot upon successful completion of testing.

TABLE IV. Special sampling.

CHARACTERISTIC	REQUIREMENT	TEST PROCEDURE
Critical:	None	N/A
Major:		
1030 Vibration	3.3.1	4.7.1.1
104. Shock	3.3.2	4.7.1.2
105. Altitude	3*3.3	4.7.1.3
106. Temperature	3.3.4	4.7.1.4
107. Humidity	3.3.5	4.7.1.5
Minor:	None	N/A

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4.5.1 Failure of sample. Should any one item of a special sampling fail to meet any of the specified test requirements, acceptance of the product shall be suspended by the Government until necessary corrections to all production since the last test as well as to current production has been accomplished in accordance with the provisions of the contract.

4.6 Inspection equipment.

4.6.1 Government-furnished inspection equipment. Where the contract provides for Government-furnished test equipment, supply and maintenance of test equipment shall be in accordance with the applicable requirements of MIL-I-45607.

4.6.2 Contractor-furnished Inspection equipment.

4.6.2.1 Government design. Unless otherwise specified in the contract, all inspection equipment identified by drawing number in specifications or QAPS forming a part of the contract shall be supplied by the contractor in accordance with the design specified. The contractor may, however, propose alternatives to Government designs for approval in accordance with 6.3.

4.6.2.2 Contractor design. The contractor shall design and supply inspection equipment compatible with the 'Methods of Inspection" specified in 4.7 of this specification and with the component inspection procedures specified in 'Examination" and 'Test Facilities" requirements of MIL-F-13926 whenever Government designs are not specified (see 4.6.2.1). Since tolerance of test equipment Is normally considered to be within 10 percent of the product tolerance for which it is intended, this inherent error in the test equipment design must be considered as part of the prescribed product tolerance limit. Thus, concept, construction, materials, dimensions and tolerances used in the design of test equipment shall be so selected and controlled as to ensure that the test equipment will reliably indicate acceptability of a product which does not exceed 90 percent of the prescribed tolerance limit and permit positive rejection when nonconforming. Construction shall be such as to facilitate routine calibration of test equipment. Contractor inspection equipment designs shall be submitted for approval in accordance with 6.3.

4.6.2.3 Test equipment. In conjunction with 4.6.2.1 and 4.6.2.2, the test equipment listed in Table V (Test equipment list) shall be used in the performance of the applicable test methods and procedures as specified in 4.7.

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TABLE V. Test equipment list.

ITEM NUMBER	DESCRIPTION
1. Oscilloscope	Dual channel vertical inputs. Alternate or individual display of each channel. Sensitivity: Adjustable 5 mV to 10 V/ division. Bandwidth: 60 MHz minimum. Time base: 50 ns/division to 50 us/division in a minimum of 10 discrete steps. Delayed sweep capability: 2 us to 50 us, continuously variable.
2. Pulse generator	Pulse amplitude: O to +15 V peak into a 50 ohm load. Pulse width: 0.1 to 5.0 us. Rise time: Not more than 25 ns.
3. Pulse generator (3 required)	Wise amplitude: O to +10 V peak into a 50 ohm load. Pulse width: 0.1 to 5.0 us. Rise time: Not more than 25 ns. Pulse delay: Variable from O to 15 us minimum.
4. Oscillator	59.936 ± 0.001 MHz, 10548158.
5. RF generator	Tuneable to 60 ± 15 MHz. output level: Adjustable from -1o7 to +10 dBm into a 50 ohm load. Harmonic content: More than 30 dB below carrier frequency.
6. Frequency counter	Accuracy: ± 1 count ± time base accuracy. Time base: variation with temperature less than ± 2 parts in 10^6 from +10° to +50°c. Variation with ± 10% line voltage plus or minus one part in 10. Range: 1 Hz to 75 MHz. Display: 6 digit readout. Sensitivity: 100 mV rms.

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TABLE V. Test equipment list (continued).

ITEM NUMBER	DESCRIPTION
7. True rms voltmeter	Range: 1 mv to 300 Vrms in 10 dB steps. Accuracy at 25°C and input frequencies of 50 Hz to 1 MHz ± l% of full scale. Input impedance: 10 megohms shunted by less than 50 pF. Response: Responds to rms value (heating value) of the input signal for all waveforms. Response time: Less than 5 seconds to respond to final value.
8. RF voltmeter	Range: 1 mV to 3 Vrms in 10 dB steps. Frequency range: 20 KHz to 100 MHz. Input impedance: 50 ohms. Accuracy: ± 3% of full scale.
9. Digital voltmeter	Range: o to 150 Vdc. Input impedance: More than 1 megohm. Readout: Digital display from O to 50 v to one decimal place. Accuracy at 25°C: ± (0.1% of reading +1 digit) .
10. Power supply	Type: Laboratory supply. Minimum requirements: O to 30 Vdc. Load regulation: 0.02%. Load current capability: 300 mA. Ripple and noise: l mV rms or 5 mV peak-to-peak maximum from DC to 20 MHz .
11. Power supply	Type: Laboratory supply. Independent dual voltage source. Minimum requirements: O to 20 vdc. Load regulation: 0.02%. Load current capability: 300 mA. Ripple and noise: l mV rms or 5 mV peak-to-peak maximum from DC to 20 MHz.
12. Power supply	Type: Laboratory supply. Minimum requirements: O to 28 Vdc. Load regulation: 0.02%. Load current capability: 100 mA. Ripple and noise: l mV rms or 5 mV peak-to-peak maximum from DC to 20 MHz.

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TABLE V. Test equipment list (continued).

ITEM NUMBER	DESCRIPTION
13. Variable attenuator (2 required)	**Attenuation: 0 to 100 dB in 1 dB steps. Frequency range: 10 to 100 MHz. Impedance input and output: 50 ohms.**
14. Hybrid junction	**10548183**
15. Balanced **mixer**	**10548166**
16. **Test harness**	**See Figure 2 and Table VI.**

TABLE VI. Test harness parts list.

DESIGNATION	PART	DESCRIPTION
P1	**Connector**	**Bendix PT06BE22-78SX(SR), or equivalent**
P2	**Connector**	Bendix PT06SE22-78SW(SR), or equivalent
P3	**Connector**	**Bendix L14TF15S6N-A, or equivalent**
P4	**Connector**	**Bendix L14TF12S6N-A, or equivalent**
P5	**Connector**	**Bendix PT06SE22-78S(SR), or equivalent**
S1	**Switch, Toggle**	**3PST**
S2	**Switch, Toggle**	DP3 Position, Center off

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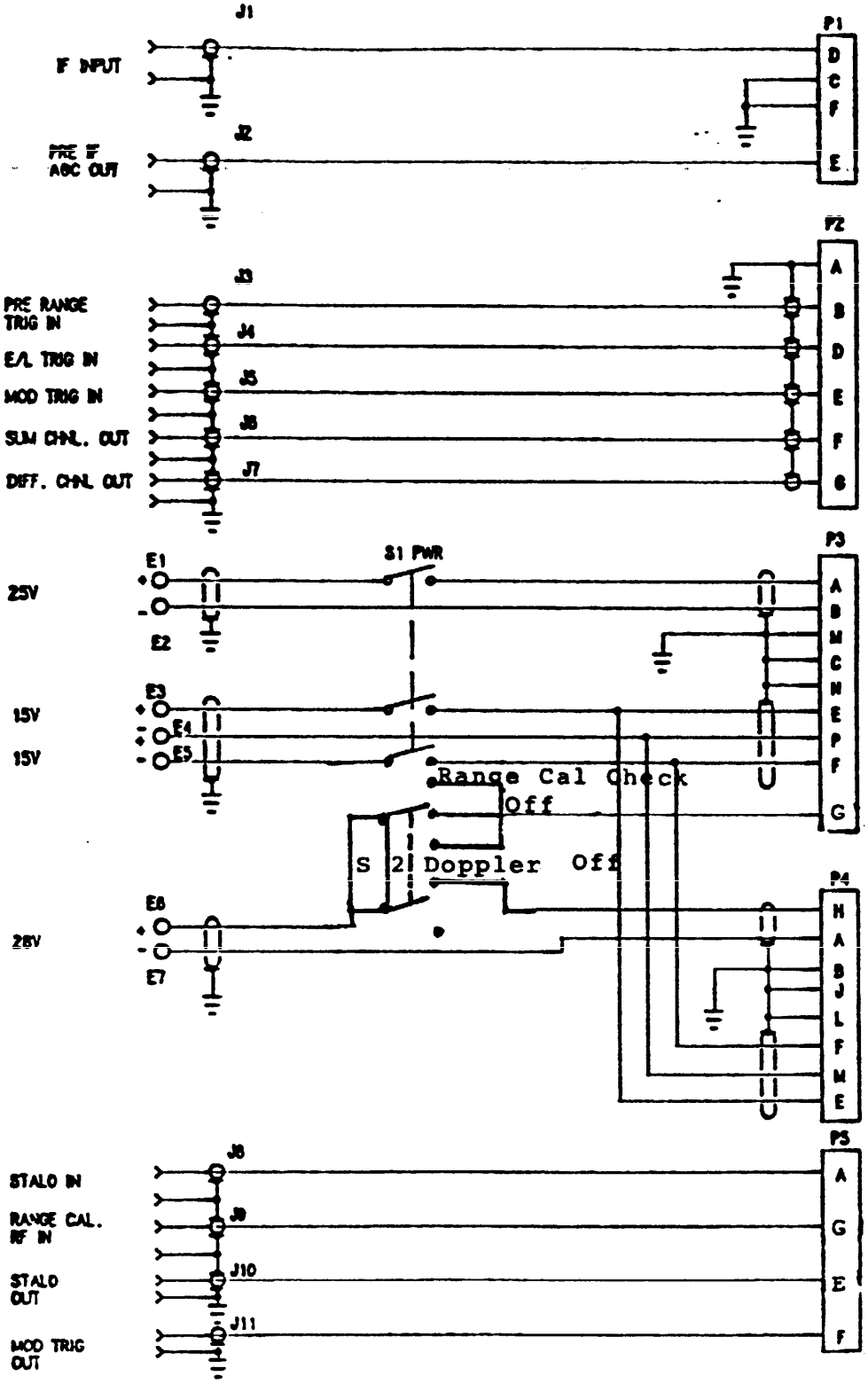


FIGURE 2. Test harness.

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TABLE VI. Test harness parts list (continued).

DESIGNATION	PART	DESCRIPTION
El through E7	Binding Post	H.H. Smith, 257
J1 through Jll	Connector, BNC	uG909/u

4.7 Methods of inspection.

4.7.1 Environmental. Environmental tests shall be conducted in accordance with the test conditions, general test performance guidance and individual test methods of MIL-STD-810, except as specified herein. Test facilities, chambers and apparatus used in conducting the tests contained in this specification shall be capable of meeting the conditions required.

4.7.1.1 Vibration. The test shall be performed as specified in Method 514, Category 8, Procedure I, of MIL-STD-810, as modified by 3.3.1 **herein. In step 1 of Procedure only "steps 4 through 8** of **paragraph II-2.1 apply. The receiver shall be attached to the vibration generator directly or with a fixture, and securely held by its normal means of attachment. After vibration, the receiver shall be tested as specified in 4.7.2 and 4.7.3 herein to determine compliance with 3.3.1.**

4.7.1.2 Shock. This test shall be performed as specified in Procedure I, Section II of Method 516 of MIL-STD-810 except that:

a. **The post test functional check (step** 5 **of Procedure I) need not be conducted until the completion of all shock impulses.**

b. **The shock pulses shall be 15 g's ± 10% amplitude and II ms ±** 10% duration.

co The maximum g shall occur at 5.5 milliseconds ± 10%.

d. **The shock measuring system shall have a flat frequency response within ± 10% from 20 to 2,000 hertz.**

After the application of all the shock impulses, the receiver shall be tested as specified in subparagraphs 4.7.2 and 4.7.3 to determine compliance with 3.3.2.

4.7.1.3 Altitude (operating and non-operating) This test shall be performed as specified in Procedures II and III, Method 500 of MIL-STD-810. The operational checkout shall be as specified in 4.7.2 and 4.7.3 for compliance with 3.3.3.

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4.7.1.4 Temperature.

4.7.1.4.1 High (operating and non-operating) This test shall be performed as specified in Procedures I and II, Method 501 of MIL-STD-810 with constant temperature extremes as specified in 3.3.4. The operational checkout shall be as specified in 4.7.2 and "4.7.3 for compliance with 3.3.4.

4.7.1.4.2 Low (operating and non-operating). This test shall be performed as specified in procedures I and II, Method 502 of MIL-STD-810 with constant temperature extremes as specified In 3.3.4. The operational checkout shall be as specified in 4.7.2 and 4.7.3 for compliance with 3.3.4.

4.7.1.5 Humidity (non-operating). This test shall be performed In accordance with Procedure III, Method 507 of MIL-STD-810 except that in step 7, checkout shall be performed at the conclusion of the test only, with the item at standard ambient conditions. The operational checkout shall be as specified in 4.7.2 and 4.7.3 for compliance with 3.3.5.

4.7.2 Electrical. Continuity shall be inspected by standard measuring equipment and visual examination for compliance with 3.4.

4.7.3 Performance. The selected receivers shall be tested as follows:

4.7.3.1 Power supply adjustment.

Using the digital voltmeter (item 9, Figure 3) to monitor the supply voltage; Set all power supply voltages as follows:

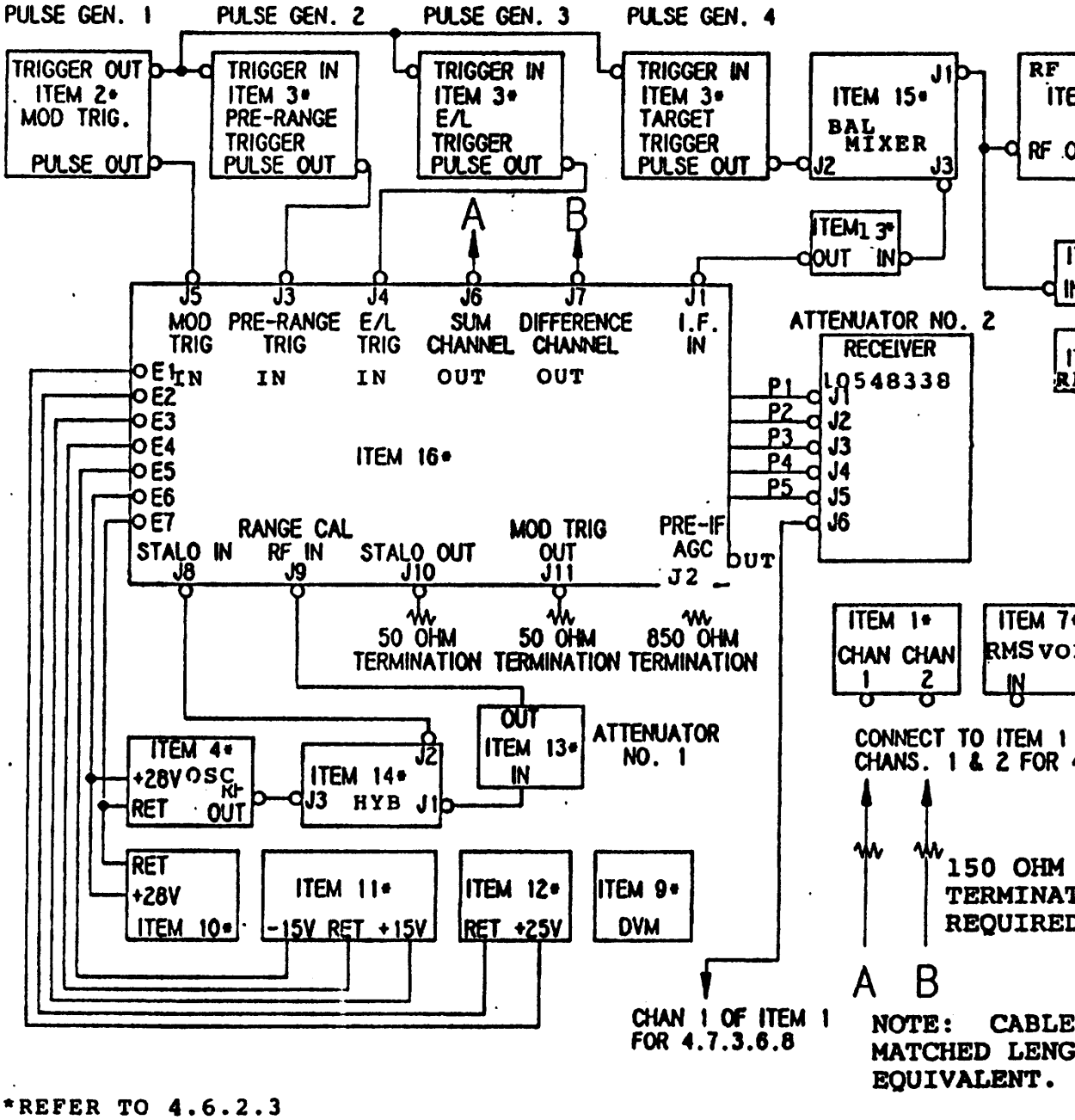
Item 10, Figure 3 - 28.0 Vdc.
b. Item 11, Figure 3 - Output No. 1, -15.0 Vdc.
c. Item 11, Figure 3 - Output No. 2, +15.0 Vdc.
d. Item 12, Figure 3 - 25.0 Vdc.

Shut off power supplies.

4.7.3.2 Pulse generator adjustment. Adjust each pulse generator (items 2 and 3, Figure 3) to produce pulses 1 through 4, respectively, as shown on the pulse timing chart, Figure 1. observe each pulse on the oscilloscope (item 1, Figure 3) with a 50-ohm termination connected at the oscilloscope input.

4.7.3.3 Test setup. Connect all test equipment as shown in Figure 3.

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*REFER TO 4.6.2.3

FIGURE 3. Receiver Test Setup.

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4.7.3.4 **Early/Late Gate RF burst amplitude.** Turn on all power supplies and S1 on the test fixture (item 16, Figure 3). Disconnect 3P42 from 3A6A5J3 and connect 3P42 to channel 1 of the oscilloscope with a 50 ohm termination at the oscilloscope input. Trigger oscilloscope with Pulse Gen. #3 output. Observe that the early and late bursts are not more than 850 mV peak-to-peak and not less than 550 mV peak-to-peak. The difference in amplitude between the two bursts shall not exceed 130 mV peak-to peak.

4.7.3.5 Sum **Gate RF burst amplitude.** Disconnect 3P45 from 3A6A6J3 and connect 3P45 to channel 2 of the oscilloscope, connecting a 50 ohm termination at the oscilloscope input. Observe that the Sum Gate RF burst occurs once each 50 microseconds (us) and that the amplitude is not less than 616 mV peak-to-peak and not more than 1.18 V peak-to-peak. Position channel 1 signal over channel 2 signal and adjust the Volts/Div variable control for equal amplitude signals. Set oscilloscope for algebraic add and adjust 322 for a null the Early Gate. Amplitude of the resultant waveform in the Early Gate period shall be no greater than one-half the sum signal amplitude. Set Channel 2 for invert and adjust 323 for a null in the Late Gate. Amplitude of the resultant waveform in the Late Gate period shall be *no* greater than one-half the sum signal amplitude.

4.7.3.6 **Early/Late and Sum Gate phasing.**

a. Connect J3 of the balanced mixer (item 15, Figure 3) to channel 1 of the oscilloscope with a 50 ohm termination at the oscilloscope. Adjust the output of the RF generator, (item 5, Figure 3), to provide a burst of 0.635 V peak-to-peak at J3 of the balanced mixer. Reconnect J3 of the balanced mixer to attenuator no. 2 (item 13, Figure 3) input. Set attenuator no. 2 for 45 dB attenuation. Set the RF generator frequency to 59.941 MHz as indicted on the frequency counter (item 6, Figure 3).

b. Connect J6 and J7 of the test fixture to channels 1 and 2 of the oscilloscope respectively (see Figure 3). Remove 3P39 from 3A1J4. Disconnect 3P65 from 3Z3J2 and connect a 50 ohm termination to both 3P65 and 3Z3J2.

c. With the oscilloscope channel selector set to alternate and synchronizing from the Sum Doppler signal (the input signal to oscilloscope channel 1), vary the delay of pulse generator no. 4 (item 3, Figure 3), to obtain a maximum audio at channel 2 on the oscilloscope. Adjust attenuator no. 2 (item 13, Figure 3) as necessary to prevent saturation of the audio signal. Adjust 322 to obtain 180° phase relationship between the signals on channel 1 and channel 2 of the oscilloscope. Reconnect 3P65 to 3Z3J2.

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d. Disconnect 3P14 from 3A5J8 and connect 50 ohm terminations to 3P14 and 3A5J8. Adjust the delay on pulse generator no. 4 (item 3, Figure 3) to obtain a maximum audio signal on oscilloscope channel 2. Adjust 3Z3 to obtain an in-phase relationship between the audio signals on channel 1 and channel 2. Reconnect 3P14 to 3A5J8.

e. Decrease the delay setting of Pulse'gener'titer no. 4 ("item 3, Figure 3) and notice that as the target burst is moved from the Late Gate to the Early Gate a slight null occurs. Move the target through this null area, using pulse generator no. 4 delay, and simultaneously adjust 3z3 for the best null.

f. First move the target into the Early Gate, and then the Late Gate. Observe that the phase relationship between oscilloscope channel 1 and channel 2 goes from 180° ± 3° to 0° ± 3°.

9" Connect J2 of the test fixture to the oscilloscope channel 1, connecting the 850-ohm termination at the oscilloscope input . Adjust the oscilloscope to obtain a good display of the AGC pulse. Disconnect 3P49 from 3A2J5 and connect an 850 ohm termination to 3A2J5. Maximize the AGC signal by adjusting the pulse generator no. 4 (item 3, Figure 3) delay until the flat portion of the pulse moves up from zero level to a maximum DC value. Verify that this DC level can be adjusted to not less than 6 v and not more than 7 V by adjusting the video gain control on 3Al.

h. Connect 3J6 on the receiver to oscilloscope channel 1. Connect 3A2J5 to oscilloscope channel 2 using a "tee" connector and an 850 ohm termination at the oscilloscope input. Connect an 850 ohm termination to J2 of the test fixture.

i. Adjust delay setting of pulse generator no. 4 (item 3, Figure 3) until the leading edge of the video pulse on oscilloscope channel 1 is 3 microseconds inside the flat of the AGC window.

j. Adjust variable attenuator number 2 (item 13, Figure 3) in 1 dB steps until the IF AGC window starts positive and then adjust RF generator (item 5, Figure 3) until the flat is +5 v. Verify that the video Input level of oscilloscope channel 1 meets the limits of 3.5.2.1.

4.7.3.7 Doppler oscillator. Connect a test cable to 3A12J2 with a "tee" connector and connect the other end of the cable to channel 1 of the oscilloscope, also connect 3P83 to the "tee" connector. Turn S2 on the test fixture to the RANGE CAL CHECK position. The audio level displayed on the oscilloscope shall meet the requirements of 3.5.3.

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4.7.3.8 **Range calibrate check.** Set attenuator no. 1, (item 13, Figure 3) for 35 dB attenuation. Turn S2 switch to the RANGE CAL CHECK position. Connect test fixture J6 to the true rms voltmeter (item 7, Figure 3) with a 150-ohm termination. Verify that the sum channel output level meets the requirements of 3.5.4.

4.7.3.9 **I-f amplifier gain.** With S2 in the RANGE CAL CHECK position, connect the output of attenuator no. 1, (item 13,Figure 3) to the RF voltmeter (item 8, Figure 3). Set the attenuator for a reading of 0 dBm (0.223 Vrms). Disconnect the voltmeter and connect the attenuator output to J9 on the test fixture. Remove 3P37 from 3A1J2 and connect the RF voltmeter to 3A1J2. The reading shall be -2 ± 2 dBm. Disconnect the voltmeter and reconnect 3P37 to 3A1J2. Remove 3P38 from 3A1J3 and connect the RF voltmeter to 3A1J3. The reading shall be the same as that at 3A1J2 ± 1 dBm.

4.7.3.10 **I-f amplifier dynamic range.** With the equipment still connected for the final reading of 4.7.3.9, reduce the attenuation in attenuator no. 1 (item 13, Figure 3) by 10 dB. The level at 3A1J3 shall Increase by not less than 9 dB. Disconnect the voltmeter from 3A1J3 and reconnect 3P38 to 3A1J3. Disconnect 3P37 from 3A1J2. Connect the RF voltmeter to 3A1J2. The reading shall increase not less than 9 dB over that obtained in 4.7.3.9. Disconnect the voltmeter, reconnect 3P37 to 3A1J2, and position test fixture switch S2 to OFF.

4.7.3.11 **Minimum discernible signal.**

4.7.3.11.1 **Minimum discernible signal, Sum channel.** Connect J6 of the test fixture to the true rms voltmeter (item 7, Figure 3) with a 150 ohm termination. Adjust the RF generator (item 5, Figure 3) and attenuator no. 2 (item 13, Figure 3) as in 4.7.3.6.1. Adjust the delay of pulse generator no. 4 (item 3, Figure 3) to obtain a maximum reading on the true rms voltmeter. Set attenuator no. 2 for 75 dB attenuation. Remove the output cable from the RF generator and note the true rms voltmeter reading. Reconnect the output cable to the RF generator. The reading shall increase not less than 3 dB.

4.7.3.11.2 **Minimum discernible signal, Difference channel.** Remove 3P19 from 3HY1J1 and install 50 ohm terminations at 3P19 and 3HY1J1 . Connect the true rms voltmeter (item 7, Figure 3) to test fixture J7. Decrease the delay of pulse generator no. 4 (item 3, Figure 3) slightly to obtain a maximum reading on the voltmeter. Disconnect the RF output cable from the RF generator (item 5, Figure 3) and note the noise level on the true rms voltmeter, reconnect the RF output cable. Note that the reading increases not less than 3 dB. Reconnect 3P19 to 3HY1J1. Disconnect 3P20 from 3HY1J2 and terminate 3P20 and 3HY1J2 with 50 ohms. Increase the delay of pulse generator no. 4 slightly until a maximum reading is obtained on the true rms voltmeter. Disconnect the RF output cable from the RF generator and note the noise level on the true rms voltmeter. Reconnect the RF output cable. The reading on the

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voltmeter shall increase not less than 3 dB. Reconnect 3P20 to 3HY1J2 and position S2 on the test fixture to OFF.

4.7.3.12 60 MHz output. Connect J10 of the test fixture to the RF voltmeter (item 8, Figure 3). The voltmeter should read not less than 1 V.

4.7.3.13 Mod trigger output. Connect test fixture J11 to channel 1 of the oscilloscope with the 50 ohm termination at the oscilloscope input. observe that the pulse display meets the requirements of 3.5.10.

4.7.3.14 System Compatibility. After performing the tests of 4.7.3.1 through 4.7.3.13, substitute the receiver into a Radar Set AN/VPS-2 and perform the system comparability tests per MIL-R-50651 as specified below. The Radar Set AN/VPS-2 and test facilities used should be as specified in MIL-R-50651.

a. Mid-range cal function (para 4.7.3.9.7)
b. SCV without AGC (para 4.7.3.9.10)
c. AGC characteristic (para 4.7.3.9.11)

5. PACKAGING

5.1 General. Preservation, packing and marking shall be in accordance with Packaging Instructions PS10548338. The level of protection shall be as specified in the procurement document.

6. NOTES

(This section. contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. The Receiver, Radar R-1475/VPS-2 is used in the Radar Set AN/VPS-2 to provide amplification of radar target signals at i-f frequency of 60 MHz, and to provide two channels of audio output for doppler shifted radar signals which are so phased as to furnish target acceleration data to the range computer used in this system when the receiver and range computer are functioning as a closed loop.

6.2 Acquisition requirements. Acquisition documents must specify the following:

a. Title, number and date of this specification.
b. Applicable stock number.
c. First Article Sample requirements.
d. Packaging requirements, if other than specified in Section 5.
e. Serialization requirements, if applicable.
f. Certificate of Conformance for each lot of shipment of product.

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9" Issue of DODISS to be cited in the solicitation, and if required, the specific issue of individual documents referenced (see 2.1.1).

6.3 Submission of contractor inspection equipment designs for approval. Submit two copies of designs as required to: Commander, U.S. Army AMCCOM, Attn: AMSMC-QAF-1, Picatinny Arsenal, NJ 07806-5000. This address will be specified on the Contract Data Requirements List, DD Form 1423 in the contract.

6.4 Drawings. Drawings listed in Section 2 of this specification under the heading U.S. Army Armament Research, Development and Engineering Center (ARDEC), may also include drawings prepared by, and identified as, U.S. Army Armament Research and Development Command (ARRADCOM), Frankford Arsenal, Rock Island Arsenal or Picatinny Arsenal drawings. Technical data originally prepared by these activities is now under cognizance of ARDEC.

6.5 Subject Term (key word) listing.

Fire control system
Doppler
Oscillator
Hybrid junction

6.6 Changes from Previous issue. Asterisks (or vertical lines) are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

Custodian:
Army-AR

Preparing activity:
Army-AR

(Project 1285 -A692)

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3. DOCUMENT TITLE

RECEIVER, RADAR R-1475/VPS-2: 10548338

4. NATURE OF CHANGE *(Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets if needed.)*

5. REASON FOR RECOMMENDATION

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a. NAME (Last, First, Middle Initial)	b. ORGANIZATION	
c. ADDRESS (Include Zip Code)	d. TELEPHONE (Include Area Code) (1) Commercial (2) AUTOVON (If applicable)	7. DATE SUBMITTED (YYMMDD)

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